ML Summer Bridge: Week 2 Study Notes

Machine Learning Class

Summer 2026

Week 2: Names, Terms and Notation

Remember that each row a dataFrame represented a different dataPoint. In our example, each row corresponds to the information about a different person. The different items we are getting information about are called **features**. The four features in our case are: father's height, mother's height, gender and height. Sometimes we may try to predict something based on our features. The item we are trying to predict is called either the **target** or the **dependent variable**. Suppose in our case we were trying to predict a person's height based on their father's height, their mother's height and their gender. In this case, we often don't list 'height' amongst the features. Instead we say 'height' is the target. It still contributes a column to our table, but it is not calld a feature. It is called a target.

If we break things up this way, we may take our current table...

Father's Height	Mother's Height	Gender	Height
70.2	60.5	M	72.545
68.4	62.2	F	69.03
73.7	64.1	M	76.265
72.5	64.4	F	72.48
74.6	68.2	F	75.45

...and turn it into two tables. One containing our features. And one that has just one column and contains our target:

Father's Height	Mother's Height	Gender
70.2	60.5	1
68.4	62.2	0
73.7	64.1	1
72.5	64.4	0
74.6	68.2	0

Height
72.545
69.03
76.265
72.48
75.45

There is code that allows us to do this in Python. While last week we covered how to access a specific column, we did not yet cover how to modify a pandas DataFrame. We are going to use the following commands:

```
y = data[['Height']]
X = data.drop(columns=['Height'])
```

The notation of using a lowercase letter y and an uppercase letter X is intentional. We won't get into the reasoning now, but it is a common convention to have 'tables' be uppercase letters and 'columns' be lowercase letters. Before we go any further, let's make a visual plot of Height first as a function of Father's Height and then as a function of Mother's Height. We will first plot an individual's height as a function of their father's height. Hit '+Code' in the top-left menu to generate a new code box. Then type the following:

```
import matplotlib as plt
data.plot(x='Father\'s Height', y='Height', kind='line')
plt.show()
```

If you run this code block by clicking the arrow at the top-left corner of the block it will generate a line graph visualizing a graph with x-axis corresponding to the father's height and y-axis corresponding to their height with lines joining the points on the graph. Then, if you click '+Code' again to generate a new code box, we can type the following to plot height as a function of mother's height.

```
data.plot(x='Mother\'s Height', y='Height', kind='line')
plt.show()
```

Notice you didn't have to write the import statement again. Now that you have some idea, you can see both the father's height and the mother's height are directly correlated with a person's height (as one would expect), but these aren't the only factors. It appears gender also plays a role.